
Beyond Drift: Stabilizing Subjective LLM Evaluation with Information-Theoretic Rubrics

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Abstract

Despite the growing use of large language models (LLMs) in subjective tasks such as role-playing, humor, emotional intelligence, and dialogue quality, their evaluation faces a pressing **reproducibility crisis**: even the same evaluator may contradict itself when re-judging the exact same sample. We attribute this instability to dimension drift, where free-form evaluation protocols (e.g., Chain-of-Thought reasoning) unpredictably shift their implicit criteria, undermining reliability. To address this fundamental challenge, we reformulate subjective evaluation as an information-theoretic optimization problem. Specifically, we propose an **Expected Information Gain (EIG)-based framework** that constructs a stable yet adaptive personalized rubric to eliminate dimension drift. Our two-stage “generate-then-score” design first produces a diverse pool of candidate evaluation questions and then selects the most informative subset via EIG, yielding explicit and repeatable criteria. Experiments on six benchmarks, including CharacterEval, The rJokes, and MT_bench, demonstrate that our approach substantially improves both evaluation consistency and alignment with human judgments, outperforming CoT-based and fixed-questionnaire baselines. These results highlight that information-theoretic questionnaire construction offers a principled and reliable path toward reproducible evaluation of subjective tasks.

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1. Introduction

Large language models (LLMs) are increasingly applied to tasks with inherently subjective dimensions, including role-playing (Tu et al., 2024), humor understanding (Narad et al., 2025), emotional intelligence (Paech, 2023), and dialogue evaluation (Ou et al., 2024; Bai et al., 2024). With their growing capabilities (Shao et al., 2026; Zhai et al., 2026), LLMs are no longer used solely as task performers, but also as evaluators in so-called “LLM-as-a-judge” frameworks, where they are expected to provide scalable, low-cost alternatives to human annotation. However, alongside this trend emerges an even more troubling phenomenon: **intra-annotator inconsistency**. The same evaluator may assign different scores, or even shift its evaluation criteria, when asked to re-judge the exact same sample. Such self-contradiction under identical conditions exposes a **reproducibility crisis**: even the same judge, faced with the same input, can contradict itself in both scores and rationales. This instability not only undermines reliability and fairness, but also cripples the ability to track genuine progress, making it a central bottleneck that threatens the comparability of research outcomes across the community.

Empirical evidence highlights how pervasive this crisis is. We identify the underlying mechanism as *dimension drift*: across repeated trials, the evaluation criteria articulated by the same judge shift unpredictably, causing judgments to vary even on identical samples. This drift stems from a fundamental weakness of free-form protocols: without a fixed rubric, evaluators can arbitrarily introduce or discard dimensions, causing unstable and inconsistent judgments. Among these protocols, chain-of-thought (CoT) reasoning in the style of G-Eval (Liu et al., 2023) is the most prominent, and strikingly, also one of the most vulnerable. As shown in Fig. 1(a), CoT exhibits lower repeatability than even a simple “Direct” method, while Fig. 1(b) provides a concrete illustration: the same judge focuses on only partially overlapping dimensions across two identical evaluations, leading to inconsistent outcomes (Zhao et al., 2023). This combination of quantitative and qualitative evidence underscores that dimension drift is not anecdotal, but systemic.

Existing approaches have tried to address this instability but remain unsatisfactory. Multi-judge and ensemble strategies

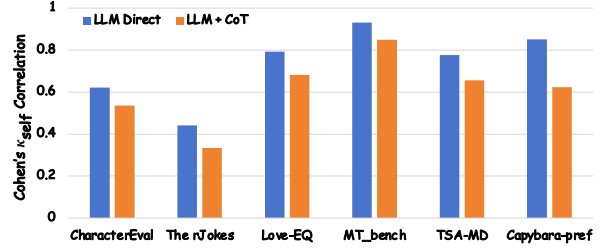
such as LMC (Zhao et al., 2024), PoLL (Verga et al., 2024), or CompassJudge (Cao et al., 2024) enhance robustness through aggregation or specialized training, but incur high computational cost and still fail to prevent drift at the individual judge level. Fixed-questionnaire methods such as CheckEval (Pereira et al., 2024) enforce consistent rubrics and thus eliminate drift, but their static design makes them brittle and insensitive to task- or sample-specific nuances. In short, the field faces a dilemma: free-form protocols (e.g., CoT) are flexible but unstable, while fixed rubrics are stable but rigid (Song et al., 2024).

How, then, can we stabilize evaluation without sacrificing task relevance? We argue for a *stability-informativeness loop*: (1) a stable rubric to eliminate dimension drift and ensure reproducibility, and (2) a principled, information-theoretic mechanism to select the most *informative* questions so that evaluations remain adaptive and task-specific. In this framing, inconsistency arises when evaluation is *rubric-blind*, while brittleness arises when it is *rubric-rigid*. We therefore introduce a generate-then-select questionnaire paradigm that unifies **stability** and **adaptivity** within a single design.

To this end, we propose an **Expected Information Gain (EIG) questionnaire framework** that reformulates subjective evaluation as an information-theoretic optimization problem. Under this view, informative evaluation questions are those whose answers maximally reduce uncertainty about the final judgment. By quantifying informativeness, stability and adaptivity are unified rather than opposed: the resulting rubric is both reproducible across trials and sensitive to task-specific nuances. Concretely, our framework (i) generates a diverse pool of candidate evaluation questions, (ii) selects a compact subset that maximizes EIG based on simulated joint distributions of answers and scores, and (iii) applies the resulting rubric to guide subsequent evaluations. This stability-informativeness loop mitigates the arbitrariness of CoT-based reasoning while avoiding the rigidity of fixed questionnaires, providing a principled solution for reproducible subjective evaluation. The source code and datasets used in our experiments are available at an anonymous repository for reproducibility: <https://github.com/xuwang0117/EIG-LLM-Eval>.

Our contributions are threefold:

- We **identify and systematically define** the problem of *dimension drift*, establishing intra-annotator inconsistency as a central bottleneck for subjective evaluation in LLM-as-a-judge settings, and introduce an explicit, task-specific questionnaire paradigm to mitigate it.
- We **reformulate subjective evaluation as an information-theoretic optimization problem**, introducing a paradigm shift that reconciles stability and



(a) Repeated-consistency on our dataset.

Analysis: The model's response did have a somewhat natural conversational tone, such as praising Zhanbo as a "good kid." However, the following topic (the lion and rabbit story) felt forced and weakly related to the characters' situation and the current conversation, and it avoided Hu Yifan's recent teasing and warning. Zeng Xiaoxian's humor and wit, his typical "cheaky and silly" radio-host persona, were not fully displayed, and the reply was overly formulaic, lacking his personalized language and emotional color. As a result, the overall reply felt mechanical and lacking in human-like interaction.

Score: 2

CoT Response 1

CoT Response 2

Analysis: From an expression perspective, the model's reply was relatively colloquial, without obvious mechanical or emotionless traits. It could naturally follow up Hu Yifan's line, and in a gentle tone, praise Zhanbo as "a good kid," showing the character's caring and softer side. At the same time, it proactively initiated a new topic (the lion and rabbit story), which added a sense of everyday life and humor, consistent with Zeng Xiaoxian's personality and human-like traits. Overall, this reply felt closer to a real person's performance.

Score: 4

(b) CoT evaluation example.

Figure 1. Repeatability analysis for LLM-as-a-judge. Panel (a) compares Direct vs. CoT under repeated evaluations on the same samples; Panel (b) shows overlap (green) and omissions/additions (red) of articulated criteria across CoT repeats.

adaptivity by selecting questions that maximize Expected Information Gain (EIG).

- We instantiate this principle in a practical stability-informativeness loop, and validate it on six benchmarks, achieving substantial gains over CoT-based and fixed-rubric baselines. In particular, our approach improves evaluation self-consistency by up to **7.6% in Cohen's κ** , while also enhancing alignment with human judgments.

2. Related Work

Evaluation Paradigms and the Stability Challenge.

Evaluating subjective tasks such as role-playing, humor, or creative writing is inherently difficult. Traditional metrics like BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), METEOR (Banerjee & Lavie, 2005), CIDEr (Vedantam et al., 2015), and embedding-based scores such as BERTScore (Zhang et al., 2019) or MoverScore (Zhao et al., 2019) capture surface overlap but miss deeper stylistic and pragmatic qualities. Human evaluation remains the "gold standard," yet is costly, slow, and inconsistent (Callison-Burch et al., 2007; Novikova et al., 2017). These challenges are especially acute in creative and open-ended tasks (Deriu et al., 2021). Against this backdrop, LLM-as-a-Judge has emerged as a scalable alternative. Models such as GPT-4, prompted with rubrics as in G-Eval (Liu et al., 2023), *sacrifice stability for adaptivity*: while flexible, they suffer from dimen-

sion drift and low repeatability. Checklist methods such as CheckEval (Lee et al., 2024) enforce *stability* but at the cost of *adaptivity* to task-dependent or instance-specific subtleties. Multi-judge ensembles such as LMC (Zhao et al., 2024), PoLL (Verga et al., 2024), or CompassJuder (Cao et al., 2024) seek robustness through aggregation, but only by increasing computational cost; each constituent judge still shifts criteria, leaving the core problem unsolved. In short, existing paradigms split between flexible but unstable protocols and stable but rigid checklists. This unresolved trade-off between stability and adaptivity motivates the need for a principled framework (Wang et al., 2025).

Rubric Design for Subjective Evaluation A complementary line of work examines how rubrics themselves are constructed. Fixed rubrics ensure consistency but often miss task-specific nuances. For example, Wei et al. (Wei et al., 2024) show that varying prompt templates or evaluation dimensions can alter both human-model agreement and retest stability, while Liu et al. (Liu et al., 2025) highlight the need to explicitly model judge consistency. These findings underscore that rubric design is not neutral: the chosen dimensions fundamentally shape evaluation outcomes. Current methods sit at two extremes: fixed rubrics provide stability but are brittle, while free-form reasoning allows adaptivity but suffers from dimension drift. Neither offers a principled way to select evaluation dimensions. In contrast, information-theoretic principles such as Expected Information Gain (EIG), widely applied in active learning (Felder & Brent, 2009) and Bayesian optimization (Frazier, 2018), provide a natural solution. Our work introduces EIG into rubric construction for subjective LLM evaluation, transforming it from a heuristic process into a principled optimization problem. This enables rubrics that are both stable across trials and adaptive to task-specific nuances, resolving the stability–adaptivity trade-off left open by prior work.

3. Methodology

We improve the reliability of subjective evaluation by explicitly materializing rubrics as questionnaires, instead of relying on implicit and unstable criteria, as shown in Figure 2. Grounded in Bayesian Experimental Design (BED), each evaluation question is treated as an experiment whose utility is measured by its expected reduction of uncertainty about the final rating y . This reformulates subjective evaluation as an *information-theoretic optimization problem*, where the goal is to select a compact set of questions that maximally reduce entropy. Our methodology proceeds in five stages: **(1) Framework and Formalism (Section 3.1)**: define evaluation instances, rating space, and the target distribution $p(y | I)$. **(2) Candidate Generation (Section 3.2)**: build a diverse, high-entropy pool Q of evaluation questions. **(3) Joint Distribution Estimation (Section 3.3)**: approxi-

mate $p(y, a | q, I)$ through LLM simulations. **(4) Expected Information Gain (Section 3.4)**: compute EIG for each question and select the most informative subset Q^* . **(5) Final Evaluation (Section 3.5)**: prompt the LLM-as-a-judge with Q^* and aggregate answers into stable ratings. The full algorithm is provided in Appendix A.2.

This stability–informativeness loop ensures that Q^* is both adaptive and reproducible: candidate diversity provides flexibility, EIG quantifies informativeness, and the fixed rubric anchors consistent evaluation, eliminating the dimension drift of free-form protocols.

3.1. Framework and Formalism

At a high level, our framework views evaluation as an information-gathering process: there exists an “ideal” score for each model response, but it is hidden. To approximate it, we ask a few structured questions whose answers gradually reduce uncertainty about the score. The key problem then becomes: which questions are most informative. This intuition motivates the formal definitions below.

Evaluation Instance: Each evaluation instance is a tuple $I = (S, M)$, where S denotes the evaluation context and M is the model’s response under evaluation. The framework is agnostic to task specifics: S may encode a role profile, a dialogue history, or any other contextual information depending on the evaluation domain.

Rating: A rating $y \in \mathcal{Y}$ (e.g., $\mathcal{Y} = \{1, 2, 3, 4, 5\}$) represents the overall subjective quality of the model response. We treat y as a latent target variable: it conceptually reflects the judgment that an ideal evaluator would assign to instance I . In practice, y summarizes diverse aspects of evaluation—such as naturalness, persona consistency, or emotional resonance—into a single scalar score. This formulation does not assume access to a ground-truth rating for every instance; rather, our framework aims to approximate this latent distribution in a stable and reproducible manner through the use of informative auxiliary questions.

Ground-truth Preference Distribution: We posit the existence of an ideal but unknown distribution $p^*(y | I)$, which represents the rating a perfect evaluator would assign to instance I . While inaccessible in practice, p^* serves as the conceptual target for reproducible and consistent evaluation.

Evaluation Question: To reduce uncertainty about y , the evaluator is guided by auxiliary evaluation questions $q \in Q$, each associated with a discrete answer space $\mathcal{A}_q$. Unlike free-form reasoning, these questions are designed to be short and structured, typically phrased as yes/no or small-scale categorical checks (e.g., {Yes, No} or {Poor, Fair, Good}). Thus, each answer $a \in \mathcal{A}_q$ is drawn from a limited, predefined set rather than open text. This discreteness makes it feasible to estimate the distribution $p(a | q, I)$ via repeated

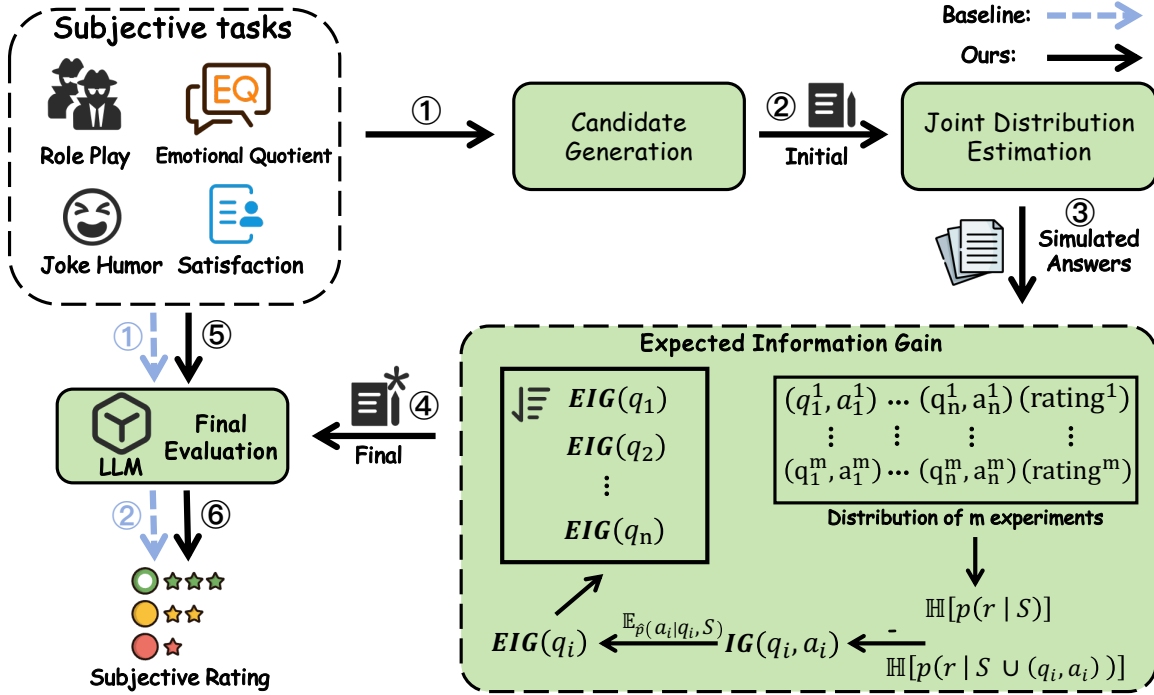


Figure 2. Overview of our framework. We construct task-specific questionnaires by generating candidate evaluation questions, estimating their informational value via simulated responses, selecting the most informative subset using Expected Information Gain (EIG), and finally conducting questionnaire-style LLM-as-a-judge evaluation.

simulations, and each (q, a) pair can be viewed as a partial observation that constrains the possible ratings, thereby providing additional information about the latent variable y .

Objective: Our framework seeks to construct a compact questionnaire $Q^* \subseteq Q$ that maximizes the expected information about y . For a question q with answer a , the information gain is defined as

$$IG(q, a) = H[p^*(y | I)] - H[p^*(y | I \cup (q, a))], \quad (1)$$

where $H[\cdot]$ is Shannon entropy. Since a is unknown beforehand, we consider the expected information gain:

$$EIG(q) = \mathbb{E}_{a \sim p(a|q, I)} [IG(q, a)]. \quad (2)$$

Intuitively, $EIG(q)$ measures how much answering q is expected to reduce our uncertainty about the final rating y . In other words, questions with higher EIG are the “most worth asking,” since their answers lead to the largest expected reduction in rating entropy. The optimized rubric is then given by

$$Q^* = \arg \max_{Q' \subseteq Q, |Q'|=k} \sum_{q \in Q'} EIG(q). \quad (3)$$

This general formalism grounds the remainder of our methodology: candidate generation defines Q , simulated responses approximate $p(y, a | q, I)$, and the final rubric Q^* emerges by optimizing for maximum expected information gain.

3.2. Constructing the Candidate Question Space

The first step is to construct the candidate space Q of evaluation questions. The goal is not to design a perfect rubric but to provide a sufficiently diverse pool for subsequent EIG-based selection. From an information-theoretic perspective, Q should approximate a high-entropy prior over evaluative dimensions, covering a wide range of response qualities while limiting redundancy. Such diversity ensures that optimization in later stages can identify the most informative subset Q^* . To build Q , we use an iterative process: the LLM is prompted with unlabeled evaluation instances to propose candidate questions (prompts in Appendix A.4), using relatively high-diversity decoding settings (temperature = 0.7, top-p = 0.9) to encourage broader exploration of potential evaluation dimensions. The generated questions are aggregated and deduplicated until the pool stabilizes. This process empirically converges to a stable region, confirming that the candidate pool quickly reaches sufficient diversity for subsequent selection (details in Section 4.3). Importantly, our framework does not require the candidate space to be manually stratified or balanced across predefined dimensions. As long as the generated pool is sufficiently diverse, the subsequent EIG optimization naturally identifies and retains the most informative questions. Consequently, candidate construction emphasizes maximizing evaluative breadth rather than enforcing handcrafted precision.

3.3. Estimating the Joint Distribution of Answers and Ratings

With the candidate pool Q constructed, the next step is to estimate the joint distribution $p(y, a \mid q, I)$, which underlies the computation of expected information gain (EIG). The detailed procedure is as follows.

Discrete Answer Space: Each evaluation question $q \in Q$ is constrained to a finite answer set $\mathcal{A}_q$ (e.g., binary, ternary, or Likert-scale). This restriction is essential: EIG requires well-defined probability distributions over answers $p(a \mid q, I)$, which are infeasible for open-ended responses. Standardizing questions into discrete formats—most often binary—keeps estimation tractable and empirically stable. We further compared binary, ternary, and 5-point scales: results show binary options achieve the most consistent evaluations, while finer granularity introduces subjectivity and drift (see Appendix A.1).

Simulated Response Filling: To approximate $p(a \mid q, I)$, we conduct independent experiments across m distinct instances I_j and questions q (Please refer to the specific prompts in Appendix A.4), yielding

$$\hat{p}(a \mid q, I) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j = a]. \quad (4)$$

Here, each I_j is a different evaluation instance, and q represents the same evaluation question posed to each instance. After simulating responses across m samples, the information provided by our joint distribution tends to stabilize (details in section 4.3).

Joint Distribution with Ratings: For each sampled answer a , the evaluator also provides a rating $y \in \mathcal{Y}$, giving

$$\hat{p}(y, a \mid q, I) = \hat{p}(a \mid q, I) \cdot p(y \mid I, q, a). \quad (5)$$

This captures how answers correlate with final ratings.

Outcome: The joint distribution $\hat{p}(y, a \mid q, I)$ enables principled computation of $EIG(q)$, linking candidate generation with information-theoretic selection.

3.4. Expected Information Gain (EIG)

We adopt the perspective of Bayesian Experimental Design (BED): an informative question is one whose answer most reduces the uncertainty of the final rating y . This uncertainty is quantified by the Shannon entropy of $p(y \mid I)$, where I is the evaluation instance.

Information Gain: For a question q_i with possible answer $a_i \in \mathcal{A}_{q_i}$, the information gain is

$$IG(q_i, a_i) = H[p(y \mid I)] - H[p(y \mid I, q_i, a_i)], \quad (6)$$

measuring the reduction in entropy once a_i is known.

Expected Information Gain: Since a_i is unknown at selection time, we compute

$$EIG(q_i) = \mathbb{E}_{a_i \sim p(a_i \mid q_i, I)} [IG(q_i, a_i)], \quad (7)$$

using empirical estimates $\hat{p}$ from the simulation procedure in Section 3.3. This makes the BED criterion operational and computable.

Optimization Objective: The final rubric Q^* is obtained by selecting k questions that maximize total informativeness:

$$Q^* = \arg \max_{Q' \subseteq Q, |Q'|=k} \sum_{q \in Q'} EIG(q). \quad (8)$$

This yields a compact, informative, and stable rubric, mitigating the drift of free-form judging.

3.5. Final Rubric Construction and Evaluation

With EIG defined, we now construct the final evaluation rubric. Given the candidate pool Q and the joint-distribution estimates from Section 3.3, we compute $EIG(q)$ for each $q \in Q$ and select a compact subset that maximizes total informativeness.

Constructing the Final Rubric: We obtain the optimized rubric by a deterministic top- k selection, following the same objective defined in Eq. 8. This procedure consolidates the outcome of prior stages into a fixed set of the most informative questions. By design, Q^* balances stability (a fixed, repeatable rubric) with adaptivity (dimensions chosen to be maximally informative for the current task domain).

Questionnaire-based Evaluation Protocol: Once Q^* is fixed, the evaluation follows a structured process where the LLM-as-a-judge assesses each question in Q^* . The LLM evaluates the responses and synthesizes them to provide a final score $\hat{y}$, reflecting the overall performance (Please refer to the specific prompts in Appendix A.4). Conceptually, $\hat{y}$ is a reproducible approximation to the latent rating y (the judgment of an ideal evaluator). Crucially, the protocol replaces free-form, ad-hoc reasoning with a pre-committed rubric, thereby eliminating dimension drift: unlike CoT-style methods that may invent or drop dimensions across runs, our judge is anchored to the same criteria every time.

Practical Considerations: Based on empirical observations, we set $k = 5$ to balance evaluation cost and key dimension coverage, keeping the rubric concise yet informative. Detailed justification is provided in Appendix A.3.

4. Experiments

4.1. Experiment Setup

Benchmarks: We evaluate our method on six benchmarks spanning both general and domain-specific subjec-

Table 1. Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)							p_{ohuman} (QWK)						
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	Avg	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	Avg
LLM Direct	0.6218	0.4420	0.7924	0.9314	0.7768	0.8519	0.7361	0.3981	0.2114	0.3203	0.4894	0.3335	0.4131	0.3610
LLM + CoT	0.5351	0.3329	0.6813	0.8494	0.6561	0.6242	0.6132	0.4617	0.2894	0.3762	0.5486	0.3994	0.4869	0.4270
PoLL	0.6317	0.4479	0.7834	0.9407	0.7714	0.8497	0.7375	0.4039	0.2210	0.3451	0.4920	0.3339	0.4474	0.3739
CheckEval	0.6398	0.4533	0.8143	0.9434	0.7821	0.8503	0.7472	0.2882	0.1902	0.2885	0.4517	0.2743	0.3781	0.3118
Ours (EIG)	0.6728	0.4832	0.8513	0.9523	0.8145	0.8974	0.7786	0.5183	0.3258	0.4203	0.5992	0.4357	0.5322	0.4719

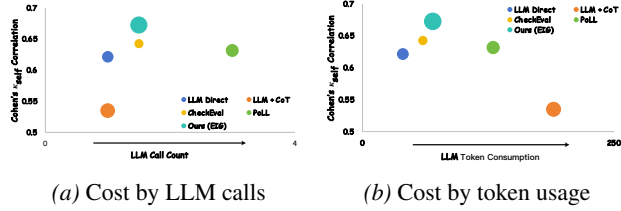


Figure 3. Cost-ability trade-offs across methods. X-axis: computational cost (LLM calls or token usage). Y-axis: self-consistency (κ_{self}). Dot size: alignment with human judgments (p_{ohuman}).

tive tasks. CharacterEval (Tu et al., 2024) measures role-playing on a 5-point Likert scale; The rJokes (Weller & Seppi, 2020) evaluates humor on a 10-point scale; Love-EQ, our new dataset, assesses emotional intelligence in romantic-chat scenarios on a 5-point scale with human annotations. MT_bench (Zheng et al., 2023) tests general response quality with binary preferences; TSA-MD (Toledo-Ronen et al., 2022) measures emotional intensity on a 5-point scale; and Capybara-pref¹ provides large-scale response preference data on a 5-point scale. **Baselines:** We compare our EIG-based framework against four strategies: LLM Direct, which scores responses without structured reasoning; LLM + CoT, which applies Chain-of-Thought prompting for guided scoring; PoLL (Verga et al., 2024), an ensemble approach that aggregates multiple LLM evaluators to improve robustness via diversity; and CheckEval (Pereira et al., 2024), which operationalizes evaluation through predefined checklists of dimensions. **Metrics:** We adopt Cohen’s κ to measure reliability of repeated evaluations, defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e the expected agreement by chance. For self-consistency, we report κ_{self} based on Cohen’s κ across repeated LLM-as-a-judge evaluations. For human alignment, we report p_{ohuman} , implemented using Quadratic Weighted Kappa (QWK), which penalizes larger rating discrepancies. **Models:** We conduct main experiments on GPT-4.1 (Achiam et al., 2023), following prior work on LLM-as-a-judge evaluation. To further test the robustness and generality of our framework across architectures and scales, we additionally evaluate with Qwen3-Next-80B-A3B-Instruct (Yang et al., 2025), Qwen3-30B-A3B-

¹<https://huggingface.co/datasets/argilla/Capybara-Preferences>

Instruct-2507 (Yang et al., 2025), DeepSeek-R1-0528 (Guo et al., 2025), Llama-3.3-70B-Instruct (Touvron et al., 2023), and Qwen3-8B (Yang et al., 2025) (see Appendix A.7). **Decoding hyperparameters:** For candidate question generation, we set temperature = 0.7, top-p = 0.9, and max tokens = 512. For simulated response filling / joint distribution estimation, final questionnaire-based evaluation, and all baseline methods, we set temperature = 0.3, top-p = 0.8, and max tokens = 512.

4.2. Overall Reliability, Cost Efficiency, and Human Alignment

We assess three critical aspects of evaluation methods: (1) *self-consistency*, (2) *alignment with human judgments*, and (3) *cost efficiency* (measured by average LLM calls and token usage).

Stability vs. Adaptivity (Table 1). On one hand, the free-form LLM + CoT method, while flexible, demonstrates the lowest stability. Its κ_{self} score is the worst among all baselines (average of 0.6132), reflecting the instability caused by *dimension drift*, i.e., shifting evaluation criteria across repeated trials. On the other hand, the **CheckEval** method exemplifies the opposite side of the trade-off: its fixed checklist yields higher self-consistency but the lowest p_{ohuman} , highlighting the brittleness of rigid rubrics that fail to capture task-specific nuances. In contrast, our proposed **EIG** framework achieves the highest scores on *both* κ_{self} and p_{ohuman} . On average, it improves self-consistency by +7.6% over CoT and +3.6% over CheckEval, while also attaining the strongest alignment with human judgments. By selecting the most informative questions, our information-theoretic rubric alleviate the stability-adaptivity dilemma: it anchors evaluations to consistent criteria (mitigating dimension drift) while remaining task-adaptive.

Cost Dimension. While Table 1 highlights performance advantages, a key motivation for LLM-as-a-judge is to offer a scalable and cost-effective alternative to human annotation. To capture this practical dimension, in our framework, we go beyond single-run cost and introduce the notion of amortized cost, defined as Amortized Cost per Sample = $\frac{\text{Upfront Cost}}{N} + \text{Per-sample Cost}$, where N is the number of

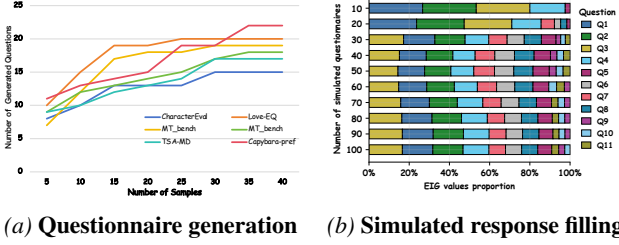


Figure 4. Effect of hyperparameters. Left: question diversity plateaus with larger sampled inputs. Right: EIG rankings stabilize with more simulated responses.

evaluated samples, *Upfront Cost* refers to the one-time expense of constructing the optimized questionnaire Q^* , and *Per-sample Cost* denotes the cost of applying it to each evaluation. Figure 3 illustrates the *cost-ability trade-off* across methods. Here, the X-axis measures amortized cost, the Y-axis reports κ_{self} , and the dot size encodes p_{ohuman} . From this perspective, LLM + CoT incurs moderate per-sample cost but exhibits the lowest stability (κ_{self}), reflecting dimension drift. CheckEval achieves higher stability yet shows weak human alignment (small dot size). PoLL delivers robustness but its amortized cost grows linearly with N , making it prohibitive for large-scale evaluations. By contrast, our EIG method requires an initial one-time investment to construct Q^* , but once established, the amortized cost per sample decreases rapidly as N grows and quickly converges to a low, stable level.

4.3. Effect of Hyperparameters

We study the sensitivity of our framework to two key hyperparameters: (i) the number of unlabeled samples used during questionnaire generation, and (ii) the number of samples used for simulated response filling. For the first hyperparameter, we examine how the diversity of generated questions evolves as more unlabeled samples are referenced. Figure 4(a) shows the average across six datasets: the number of unique (deduplicated) questions grows but quickly plateaus, indicating that a sufficiently high-entropy candidate pool can be obtained with a reasonable sample size. This ensures that EIG-based selection operates over a diverse yet stable set of evaluative dimensions. For the second hyperparameter, we analyze how the stability of question selection changes with more simulated answers. Figure 4(b) illustrates one CharacterEval role: normalized EIG values gradually stabilize as the number of simulated responses increases, showing that question importance rankings converge once the sample size is large enough. This convergence anchors the final rubric to reproducible criteria rather than stochastic fluctuations. Together, these results empirically support our **stability-informativeness closed loop**: candidate diversity guarantees adaptivity, while EIG-based stabilization secures consistency, jointly suppressing

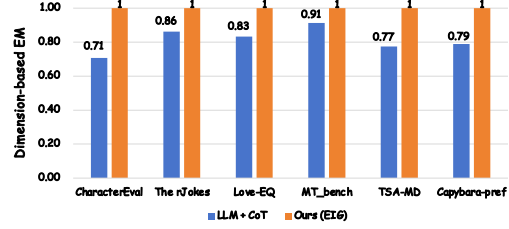


Figure 5. Dimension drift measured by EM_{dim} . LLM + CoT shows mild drift on simpler tasks (e.g., MT_bench) and much stronger drift on complex ones (e.g., CharacterEval).

the dimension drift that undermines free-form protocols.

4.4. Coverage of Candidate Questions

Beyond maximizing informativeness, it is also important to verify whether the candidate questions selected by EIG sufficiently cover the key evaluation dimensions of each task, rather than overfocusing on only a few highly discriminative aspects. In our framework, coverage is first encouraged during the candidate question generation stage. As described in Sec. 3.2, we use temperature = 0.7 to encourage the LLM to generate candidate questions from more diverse perspectives. We further combine multi-round generation, aggregation, and semantic deduplication to reduce redundancy while preserving broad evaluation viewpoints. Empirically, we also observe that the number of discovered candidate questions gradually converges as more samples are used for question generation (Figure 4(a)), suggesting that the candidate question pool can stably cover the major evaluation dimensions of the task.

To further examine whether the candidate questions selected by EIG cover the key evaluation dimensions, we additionally conduct a human-aligned dimension recall analysis. Specifically, we invite four human annotators to independently summarize two sets of core evaluation dimensions for each benchmark task: a compact set containing 5 core dimensions and a broader set containing 10 core dimensions. These dimensions are designed to jointly reflect both importance and coverage breadth. We then compare the coverage achieved by the Top-5 / 10 candidate questions selected by EIG against randomly selected sets of 5 / 10 candidate questions.

We define coverage as the proportion of human-identified core dimensions that are covered by the selected candidate questions, and report the average results across the four annotators. Table 2 shows that EIG-selected candidate questions consistently achieve substantially higher coverage than random selection across all six benchmarks. In particular, the Top-5 questionnaire selected by EIG achieves an average coverage of 89.17%, compared with only 60.0% for random selection. When increasing the questionnaire size to Top-10,

Table 2. Human-aligned coverage of EIG-selected candidate questions across six benchmarks. Coverage is measured as the proportion of human-identified core dimensions covered by the selected candidate questions.

Dataset	EIG-5	Rand-5	EIG-10	Rand-10
CharEval	90.0%	60.0%	92.5%	80.0%
rJokes	85.0%	55.0%	92.5%	72.5%
Love-EQ	100.0%	60.0%	95.0%	75.0%
MT _{bench}	90.0%	65.0%	90.0%	75.0%
TSA-MD	85.0%	60.0%	97.5%	72.5%
Capybara-pref	85.0%	60.0%	97.5%	75.0%
Avg	89.17%	60.0%	94.17%	75.0%

EIG further improves the average coverage to 94.17%, while random selection reaches 75.0%. These results suggest that although our objective is information-theoretic selection via EIG, the resulting candidate question sets do not sacrifice coverage.

4.5. Dimension-based Exact Match (EM).

To quantitatively analyze the dimension drift in Figure 1(b), we introduce the dimension-based exact match metric (EM_{dim}). An LLM first extracts a set of evaluation dimensions $G = \{g_1, g_2, \dots, g_m\}$ from the initial subjective evaluation. Then G and the re-test evaluation are provided to the LLM, which outputs the proportion of dimensions in G preserved in the second response.

Formally, $EM_{dim} = k/|G|$, where $|G|$ is the number of extracted dimensions and k the subset preserved in the second response. This measures how well initial evaluation dimensions are retained in re-testing. As shown in Figure 5, LLM + CoT analyses exhibit mild drift on simpler tasks (e.g., **MT_{bench}**, **The rJokes**) but more pronounced drift on complex ones (e.g., **CharacterEval**, **TSA-MD**). By contrast, EIG achieves $EM_{dim} = 1.0$ across all tasks since the fixed questionnaire Q^* eliminates drift. Importantly, Table 1 shows that EIG also reaches the highest human alignment (p_{ohuman}). Thus, unlike rigid checklist baselines such as CheckEval, EIG combines *stability* (no drift) with *adaptivity* (task-relevant dimensions), alleviating the stability–adaptivity dilemma discussed in the introduction.

Finally, EM_{dim} should be understood as an approximate measure whose reliability depends on the LLM. Its value lies not in providing absolute statistics, but in offering a quantitative perspective that captures *relative trends* of dimension drift across methods.

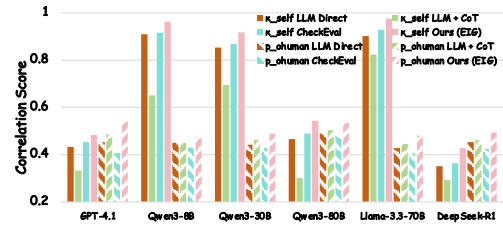
4.6. Robustness Experiment

We evaluate robustness along two axes: (i) model scale and questionnaire construction; (ii) heterogeneous generator–evaluator pipelines for cross-model transferability.

4.6.1. ROBUSTNESS TO MODEL SCALE AND QUESTIONNAIRE CONSTRUCTION

We first examine robustness under variations in foundation model scale and questionnaire construction strategies. Figure 6a reports κ_{self} and p_{ohuman} for different foundation models on the rJokes dataset. For Qwen3, κ_{self} decreases as model size increases, while p_{ohuman} improves. This trend likely reflects that larger models exhibit more divergent reasoning patterns while maintaining stronger alignment with human judgment. Similar behaviors are observed for DeepSeek-R1, Llama-3.3, and GPT-4.1, indicating that this phenomenon is consistent across model families rather than being model-specific.

Figure 6b further compares three questionnaire construction strategies: *All candidate questions*, *Random k questions*, and *Ours*. Both baselines underperform our method, but in different ways. The *All* strategy achieves higher p_{ohuman} , suggesting stronger alignment with human preferences, whereas *Random* yields higher κ_{self} , indicating stronger internal consistency. Our approach achieves a more balanced trade-off between these two metrics and consistently outperforms both baselines.



(a) Different foundation models’ performance



(b) Performance of different final questionnaire selection methods

Figure 6. Robustness analysis. Left: κ_{self} and p_{ohuman} across models on The rJokes. Right: comparison of questionnaire selection strategies.

4.6.2. ROBUSTNESS TO HETEROGENEOUS MODEL PIPELINES

We further evaluate robustness under *heterogeneous model pipelines*, where the question generator (EIG estimator) and the evaluator are instantiated with different foundation models (Table 3). Specifically, we decouple these two roles and permute them across six representative models, including GPT-4.1, Llama-3.3-70B, Qwen3-8B/30B/Next-80B, and

Table 3. Heterogeneous-variant experiments using different LLMs in the two stages of our framework—EIG estimation and questionnaire-based evaluation—comparing self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)							p_{ohuman} (QWK)						
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	Avg	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	Avg
G (baseline best)	0.6398	0.4533	0.8143	0.9434	0.7821	0.8519	0.7472	0.4974	0.3061	0.5126	0.4133	0.4238	0.5300	0.4472
G + G	0.6728	0.4832	0.8513	0.9523	0.8145	0.8974	0.7786	0.5186	0.3394	0.5455	0.4533	0.4367	0.5925	0.4810
Q8 + G	0.6642	0.4817	0.8561	0.9573	0.8121	0.8836	0.7758	0.5103	0.3346	0.5438	0.4512	0.4352	0.6025	0.4796
Q30 + G	0.6761	0.4803	0.8539	0.9623	0.8127	0.8971	0.7804	0.5036	0.3274	0.5342	0.4483	0.4248	0.5987	0.4728
Q80 + G	0.6747	0.4772	0.8368	0.9444	0.8095	0.8924	0.7725	0.5078	0.3442	0.5405	0.4669	0.4417	0.5823	0.4806
L + G	0.6583	0.4792	0.8428	0.9531	0.8047	0.8789	0.7695	0.5122	0.3310	0.5482	0.4424	0.4381	0.5849	0.4761
D + G	0.6745	0.4624	0.8526	0.9588	0.8120	0.8912	0.7753	0.5034	0.3338	0.5329	0.4512	0.4411	0.5891	0.4753
Q80 (baseline best)	0.5713	0.4892	0.7236	0.8591	0.6637	0.6484	0.6592	0.5213	0.3031	0.4751	0.4303	0.4651	0.5543	0.4582
Q80 + Q80	0.6092	0.5426	0.7949	0.8963	0.7024	0.6852	0.7051	0.5587	0.3344	0.4902	0.4894	0.4892	0.5884	0.4917
Q8 + Q80	0.6142	0.5368	0.7903	0.8912	0.7003	0.6902	0.7038	0.5466	0.3293	0.4857	0.4744	0.4742	0.5734	0.4806
Q30 + Q80	0.5987	0.5306	0.7899	0.8835	0.6904	0.6749	0.6947	0.5491	0.3447	0.5102	0.4881	0.5053	0.6081	0.5009
L + Q80	0.6003	0.5431	0.7942	0.9143	0.7013	0.6732	0.7044	0.5672	0.3312	0.4952	0.4845	0.4885	0.5957	0.4937
D + Q80	0.5910	0.5417	0.7874	0.8932	0.6954	0.6821	0.6985	0.5531	0.3249	0.5027	0.4829	0.4847	0.5841	0.4887
G + Q80	0.6036	0.5429	0.7893	0.9032	0.6948	0.6935	0.7046	0.5581	0.3317	0.4846	0.4721	0.4821	0.5912	0.4866

DeepSeek-R1.

Across all heterogeneous configurations, both self-consistency (κ_{self}) and human alignment (p_{ohuman}) remain highly stable compared to homogeneous baselines, indicating that the evaluation dimensions extracted by EIG generalize well across model families. For example, when GPT-4.1 serves as the evaluator, replacing the homogeneous setup (GPT-4.1 + GPT-4.1) with a heterogeneous one (Qwen3-Next-80B + GPT-4.1) results in nearly identical performance in terms of p_{ohuman} (0.4810 vs. 0.4806) and κ_{self} (0.7786 vs. 0.7725). Moreover, heterogeneous configurations can even outperform homogeneous ones: when Qwen3-Next-80B is used as the evaluator, employing Llama-3.3 as the question generator achieves higher human alignment than the homogeneous Qwen3-Next-80B baseline. These results demonstrate that EIG captures task-intrinsic evaluation dimensions that are transferable across models with different architectures and training objectives.

5. CONCLUSION

We identified dimension drift as a core obstacle to reproducible LLM-based subjective evaluation and proposed an EIG-driven rubric framework to resolve it. By selecting the most informative evaluation questions, our method achieves both stability and adaptivity in a unified design. Experiments across six benchmarks show consistent gains in self-consistency, human alignment, and cost efficiency over existing baselines. This establishes information-theoretic rubric construction as a principled foundation for scalable and trustworthy LLM-as-a-judge systems.

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Impact Statement

This paper presents an information-theoretic framework for improving the reliability and reproducibility of subjective LLM evaluation. The work is primarily methodological, aiming to strengthen evaluation practices rather than introduce new model capabilities. By promoting more stable and transparent benchmarking, the proposed approach may improve the comparability of experimental results and reduce evaluation noise in subjective tasks.

Several limitations should also be noted. The framework remains dependent on the underlying LLM used for question generation, simulation, and evaluation, meaning that evaluator biases may propagate into the constructed rubric. In addition, using LLM simulations to optimize an LLM-based judge may partially reflect the evaluator model’s own inductive biases. The external validity of the Love-EQ dataset may also be limited by annotation scope and domain-specific assumptions. Potential negative societal impacts should also be considered. More stable automated evaluation may encourage increased reliance on LLM-based judges in socially sensitive applications.

Overall, we believe this work contributes toward more rigorous and transparent LLM evaluation, while highlighting the need for further research on evaluator bias and human-centered validation.

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A. Appendix

A.1. Effect of Response Granularity on EIG Stability

Section 1 identified dimension drift as the main source of instability in free-form evaluation. Our EIG framework mitigates this by fixing a questionnaire, suppressing macro-level drift. A natural question is whether the answer format itself introduces instability at a finer scale. We test this on CharacterEval by holding the question set fixed and varying answer granularity: binary (yes/no), ternary, and 5-point Likert (Fig. 7). Results show a clear trade-off: (i) with more options, absolute EIG differences shrink; (ii) however, question rankings by EIG become less stable, as finer-grained scales amplify LLM subjectivity; (iii) binary options yield the most consistent rankings. Thus, response granularity introduces a form of micro-level drift. To build a robust evaluation system, both informative question selection and stable answer formats are necessary. Binary choices, beyond computational simplicity, are a key design choice for ensuring end-to-end stability.

A.2. Algorithmic Details of Questionnaire Construction and Evaluation

To ensure transparency and reproducibility, we provide the full pseudocode for our Expected Information Gain (EIG)-based rubric construction. This algorithm instantiates the Bayesian Experimental Design perspective described in Section 3, where evaluation questions are treated as experiments whose utility is measured by the expected reduction in uncertainty about the latent rating.

The procedure begins by generating a diverse pool of candidate evaluation questions Q using LLM proposals on unlabeled instances. Each question is standardized into a discrete answer space $\mathcal{A}_q$ (e.g., binary or small categorical options), ensuring tractable probability estimation and eliminating duplicates. For each question–instance pair, we approximate the answer distribution $\hat{p}(a \mid q, I)$ via repeated LLM simulations, and conditionally elicit corresponding ratings $p(y \mid I, q, a)$. Together, these yield an empirical joint distribution $\hat{p}(y, a \mid q, I)$, which anchors the subsequent information-theoretic computation.

We then compute the Expected Information Gain (EIG) of each candidate question:

$$\text{EIG}(q) = \frac{1}{|\mathcal{D}|} \sum_{I \in \mathcal{D}} \sum_{a \in \mathcal{A}_q} \hat{p}(a \mid q, I) \cdot \left(H[p(y \mid I)] - H[p(y \mid I, q, a)] \right).$$

Intuitively, questions with higher EIG values contribute more to reducing entropy in the rating distribution, and are thus more informative for stabilizing evaluation. The final rubric Q^* is obtained via a top- k selection that maximizes

Algorithm 1 Detailed Procedures of Questionnaire Construction and Evaluation

Require: Evaluation instances $\mathcal{D} = \{I = (S, M)\}$, rating space $\mathcal{Y}$, simulation rounds m , target size k
Ensure: Final rubric Q^* and reproducible rating $\hat{y}$

- 1: Generate candidate questions Q via LLM proposals on $\mathcal{D}$
- 2: Standardize each $q \in Q$ into a discrete answer space $\mathcal{A}_q$ and remove duplicates
- 3: **for** each $q \in Q$ and $I \in \mathcal{D}$ **do**
- 4: Estimate $\hat{p}(a \mid q, I)$ by m -round simulation
- 5: **for** each $a \in \mathcal{A}_q$ **do**
- 6: Obtain conditional distribution $p(y \mid I, q, a)$
- 7: Compute $\hat{p}(y, a \mid q, I) \leftarrow \hat{p}(a \mid q, I) \cdot p(y \mid I, q, a)$
- 8: **end for**
- 9: **end for**
- 10: **for** each $q \in Q$ **do**
- 11: Compute $\text{EIG}(q) \leftarrow \frac{1}{|\mathcal{D}|} \sum_{I \in \mathcal{D}} \sum_{a \in \mathcal{A}_q} \hat{p}(a \mid q, I) \left(H[p(y \mid I)] - H[p(y \mid I, q, a)] \right)$
- 12: **end for**
- 13: Select final rubric $Q^* \leftarrow \arg \max_{Q' \subseteq Q, |Q'|=k} \sum_{q \in Q'} \text{EIG}(q)$
- 14: For evaluation: ask LLM-as-a-judge each $q \in Q^*$ on instance I
- 15: Aggregate answers $\{a_q\}_{q \in Q^*}$ into final score $\hat{y}$
- return** $Q^*, \hat{y}$

the total informativeness across all chosen questions.

During evaluation, the LLM-as-a-judge is constrained to answer only the rubric questions in Q^* , producing structured answers $\{a_q\}_{q \in Q^*}$. These answers are then aggregated into a reproducible rating $\hat{y}$ through a deterministic mapping, ensuring stability and eliminating dimension drift.

The complete procedure is summarized in Algorithm 1.

This pseudocode operationalizes the stability–informativeness loop described in the main text: candidate diversity enables adaptivity, EIG quantifies informativeness, and the final rubric enforces stability across repeated evaluations.

A.3. Number of Questions k

Based on experiments conducted on six datasets, the **bars** in the figure represent the average Cohen’s κ_{self} correlation scores, while the **line** represents the average p_{ohuman} QWK scores. It can be observed that κ_{self} remains relatively stable as the number of questions k increases, but starts to decline when $k > 5$. In contrast, p_{ohuman} first increases with larger k , and then gradually decreases. Therefore, we set $k = 5$ as the parameter for our main experiments.

A.4. Prompt Template for Questionnaire Generation

In this subsection, we present the prompt templates used for generating candidate questions, simulating questionnaire

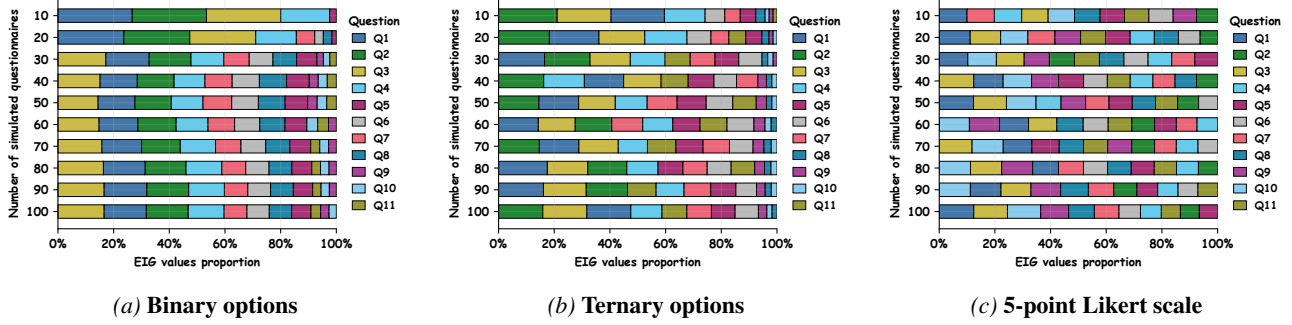


Figure 7. Effect of response granularity on EIG. Simulated responses with increasing sample size show how expected information gain (EIG) changes under binary, ternary, and 5-point Likert settings.

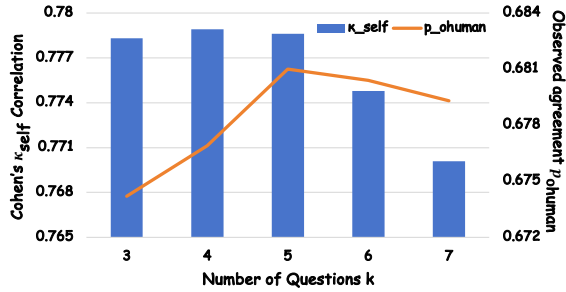


Figure 8. Results of LLM Direct, LLM CoT, and Ours on human consistency metrics.

responses and final evaluation.

A.5. Effect of Deterministic Decoding

Table 4 reports the effect of deterministic decoding on CharacterEval and MT-bench. We evaluate LLM Direct, LLM + CoT, PoLL, CheckEval, and our EIG-based framework using GPT-4.1 under two decoding settings: temperature = 0.3 and temperature = 0. Across both datasets, setting temperature to 0 consistently improves self-consistency (κ_{self}), indicating that deterministic decoding can effectively reduce randomness introduced by sampling. However, human alignment (p_{ohuman} , measured by QWK) slightly decreases in most cases, suggesting that more deterministic outputs do not necessarily lead to stronger agreement with human judgments.

These results indicate that temperature = 0 mainly mitigates decoding-level randomness, but does not fully address the core issue studied in this work, namely dimension drift, where the evaluator changes its implicit evaluation criteria across repeated assessments of the same input. In contrast, our method stabilizes the evaluation criteria themselves through a fixed and informative rubric, targeting criterion-level instability rather than only sampling variance. Therefore, deterministic decoding is a useful complementary setting, but it cannot replace the need for explicit

Prompt template for Candidate Generation

System Prompt

As a role-playing dialogue evaluation expert, please generate a series of "Yes/No" format evaluation questions based on the provided character information, evaluation dimensions, and reference samples. The aim is to comprehensively assess the dialogue quality of the character.

Instruction

<Role Profile>: 'Name': Tong Xiangyu; 'Identity': Owner of Tongfu Inn, daughter of the Longmen Escort Agency; 'Personality': Selfish, greedy, vain, and concerned with appearances, yet also seeks enjoyment; 'Background': Originally set to marry the head of Mount Heng Sect, she became a widow on the way. She ...

<Reference Samples>: Tong Xiangyu: (playfully whining) "Oh, come on! Such a simple question, and you need so long to think?"
Bai Zhanlang: "Can we change the question?"

...
Tong Xiangyu: "Hmph, so funny. If you have such a great sense of humor, why don't you go write a sitcom?"
Bai Zhanlang: "I still prefer careers with ambition, like being a waiter."

<Task Requirements>: Please generate a series of "Yes/No" evaluation questions based on the specific information and evaluation dimensions for the character. The specific requirements are as follows:

1. Each question should be in the "Yes/No" format, with answers being "Yes" or "No."
 2. All questions should cover various aspects of the evaluation dimensions and be designed based on the character's specific traits and background.
 3. The questions should be clear, specific, and objective, making it easy for evaluators to make judgments.
 4. The question design should closely align with the character's traits, reflecting their performance in dialogue.
 5. Each question should focus on one evaluation point, avoiding lengthy or multi-part questions.
- Please output the list of questions directly, one question per line.

Figure 9. Prompt for Generating Candidate Questions

rubric stabilization. Notably, our method achieves the best self-consistency under temperature = 0 and the strongest human alignment under temperature = 0.3 on both datasets, demonstrating its effectiveness under both stochastic and deterministic decoding settings.

A.6. Fair Comparison under Multi-Round Score Aggregation

Multi-round inference followed by score aggregation is a simple, practical, and comparable strategy for improving evaluation stability. However, this setting differs from the stage studied in our work. Specifically, multi-round aggregation is typically a post-processing mechanism applied after repeated evaluations, whereas our method focuses on stabilizing the evaluation criteria themselves during the evaluation process. To enable a fair comparison, we therefore treat multi-round aggregation as a unified post-processing

Table 4. Effect of deterministic decoding on CharacterEval and MT-bench. We compare self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK) under temperature = 0.3 and temperature = 0. Best results are in bold, second-best are underlined.

Method	CharacterEval κ_{self}	CharacterEval p_{ohuman}	MT-bench κ_{self}	MT-bench p_{ohuman}
LLM Direct (T=0.3)	0.6218	0.3981	0.9314	0.4894
LLM Direct (T=0)	0.8476	0.3974	0.9821	0.4779
LLM + CoT (T=0.3)	0.5351	0.4617	0.8494	0.5486
LLM + CoT (T=0)	0.7429	0.4431	0.9746	0.5378
PoLL (T=0.3)	0.6317	0.4039	0.9407	0.4920
PoLL (T=0)	0.8657	0.3997	<u>0.9838</u>	0.4809
CheckEval (T=0.3)	0.6398	0.2882	0.9434	0.4517
CheckEval (T=0)	<u>0.8791</u>	0.2796	0.9831	0.4502
Ours (EIG) (T=0.3)	<u>0.6728</u>	0.5183	0.9523	0.5992
Ours (EIG) (T=0)	0.8965	<u>0.4971</u>	0.9934	<u>0.5886</u>

Prompt template for Simulated Response Filling

System Prompt

As a role-playing dialogue evaluation expert, you will conduct a detailed assessment of the provided role-playing dialogue sample. Please answer each question based on the content of the dialogue, the character profile, and the evaluation questionnaire, and provide an overall score based on your answers.

Instruction

<Role Profile>: 'Name': Tong Xiangyu; 'Identity': Owner of Tongfu Inn, daughter of the Longmen Escort Agency; 'Personality': Selfish, greedy, vain, and concerned with appearances, yet also seeks enjoyment; 'Background': Originally set to marry the head of Mount Heng Sect, she became a widow on the way. She ...

<Reference Samples>: Tong Xiangyu: (playfully whining) "Oh, come on! Such a simple question, and you need so long to think?"
Bai Zhantang: "Can we change the question?"

...
Tong Xiangyu: "Hmph, so funny. If you have such a great sense of humor, why don't you go write a sitcom?"
Bai Zhantang: "I still prefer careers with ambition, like being a waiter."

<Task Requirements>

1. Please answer each question objectively and accurately, based on the character profile and the content of the dialogue.
2. For each question, pay special attention to whether the character matches the traits as defined in their profile and how they are portrayed in the dialogue.
3. The final score should consider the performance across all evaluation questions in the dialogue.

Please strictly follow the format below for your response:

Question 1: Yes/No

...

Question {len(questions)}: Yes/No

Overall score: X points

Prompt template for Final Evaluation

System Prompt

As a role-playing dialogue evaluation expert, you will conduct a detailed assessment of the provided role-playing dialogue sample. Please answer each question based on the content of the dialogue, the character profile, and the evaluation questionnaire, and provide an overall score based on your answers.

Instruction

<Role Profile>: 'Name': Tong Xiangyu; 'Identity': Owner of Tongfu Inn, daughter of the Longmen Escort Agency; 'Personality': Selfish, greedy, vain ...

<Evaluated Samples>: Tong Xiangyu: (playfully whining) "Oh, come on! Such a simple question, and you need so long to think?"
Bai Zhantang: "Can we change the question?"

...

<Evaluation Questionnaire>

1. Does the character's response reflect their personality traits (e.g., quirky, lively, straightforward, etc.)?
2. Does the speaking style align with their historical or martial arts background and life experiences?

...

<Task Requirements>

1. Base your responses strictly on the evaluation questions from the questionnaire; do not provide any additional reasoning or explanations.
2. Maintain objectivity and neutrality, without any subjective opinions.
3. Only output the numeric score, without including any other form of response.
4. Each question should be answered based on the character's traits and background, strictly evaluating their performance in dialogue.

Respond in this format:

Score: [1-5]

Figure 10. Prompt for Simulating Questionnaire Responses

Figure 11. Prompt for Simulating Questionnaire Responses

mechanism and apply it consistently to both our method and all baseline methods.

Table 5 reports the effect of multi-round score aggregation on CharacterEval and MT-bench. We use GPT-4.1 as the evaluator and compare single-run evaluation with a 3-round aggregation scheme, where the final score is obtained by averaging the scores from three independent runs and rounding the result to the nearest integer. This aggregation strategy is applied uniformly to LLM Direct, LLM + CoT, PoLL, CheckEval, and our EIG-based framework.

The results show that 3-round aggregation consistently improves both self-consistency (κ_{self}) and human alignment (p_{ohuman} , measured by QWK) across all methods. This confirms that multi-round aggregation is an effective post-

processing strategy for reducing evaluation variance. Importantly, under the same aggregation mechanism, our method still achieves the best performance on both datasets in terms of both self-consistency and human alignment. These results indicate that the advantage of our approach does not merely arise from repeated inference or aggregation itself, but from the improved stability and stronger human alignment of the underlying evaluation mechanism.

A.7. The Performance of Different Foundation Models

Tables 6–10 report the performance of our method and baseline evaluators across different foundation models, including Qwen3-8B, Qwen3-30B-A3B-Instruct-2507, Qwen3-Next-80B-A3B-Instruct, Llama-3.3-70B-Instruct, and DeepSeek-R1-0528. For each model, we compare self-consistency

Table 5. Comparison of single-run evaluation and 3-round average-then-round aggregation on CharacterEval and MT-bench. We compare self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are underlined.

Dataset	Method	Single-run κ_{self}	3-round Avg+Round κ_{self}	Single-run p_{ohuman}	3-round Avg+Round p_{ohuman}
CharacterEval	LLM Direct	0.6218	0.6412	0.3981	0.4013
	LLM + CoT	0.5351	0.5768	<u>0.4617</u>	<u>0.4642</u>
	PoLL	0.6317	0.6495	0.4039	0.4068
	CheckEval	<u>0.6398</u>	<u>0.6534</u>	0.2882	0.2911
	Ours (EIG)	0.6728	0.6907	0.5183	0.5207
MT-bench	LLM Direct	0.9314	0.9389	0.4894	0.4916
	LLM + CoT	0.8494	0.8816	<u>0.5486</u>	<u>0.5513</u>
	PoLL	0.9407	0.9462	0.4920	0.4946
	CheckEval	<u>0.9434</u>	<u>0.9471</u>	0.4517	0.4539
	Ours (EIG)	0.9523	0.9596	0.5992	0.6018

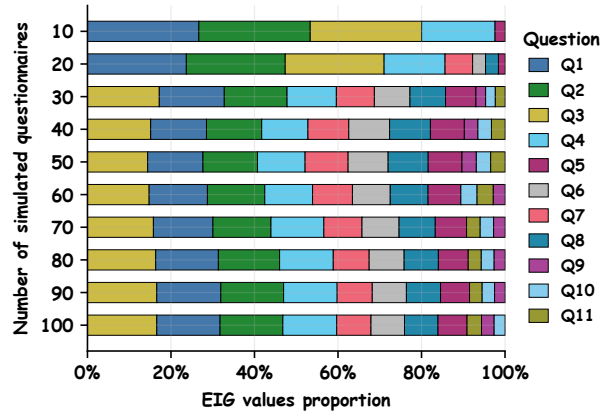
(κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK), with best results highlighted in bold and second-best results shaded. Across all foundation models, our EIG-based framework consistently achieves the highest self-consistency and strongest alignment with human judgments, demonstrating that the proposed information-theoretic rubric construction is robust to model scale, architecture, and training paradigm. While absolute performance levels vary across models, the relative trends remain stable: free-form baselines exhibit lower consistency, whereas our method reliably improves both reproducibility and human alignment, indicating strong cross-model generalization of the selected evaluation dimensions.

A.8. Love-EQ Annotation Guidelines and Procedure

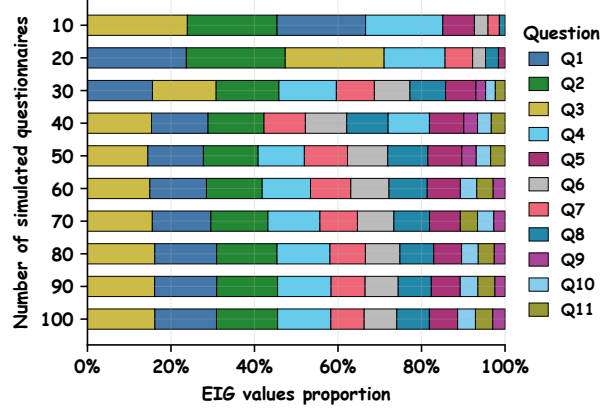
A.8.1. ANNOTATION GUIDELINE

This guideline was provided to all annotators participating in the construction of the Love-EQ dataset. The annotation aims to assess the overall dialogue quality and emotional intelligence (EQ) exhibited by the *male speaker* in romantic/flirting chat scenarios. Specifically, for each dialogue, annotators are asked to assign a **single holistic score** on a **5-point scale (1–5)** to the male utterance marked as “Golden”, after considering the full conversational context.

- **Task:** Given a multi-turn romantic/flirting conversation (with both male and female utterances), annotators should **focus only on the male “Golden” utterance**. Using the full context, annotators judge whether the male speaker’s response is natural, appropriate, humorous/clever when applicable, and emotionally attuned. Annotators do *not* score each dimension separately; instead, they should weigh the following dimensions as references and finally provide one overall quality score (1–5).
- **Reference dimensions (for guidance only; no per-dimension scoring required):**



(a) Uniform random sampling from D .



(b) Cluster-then-stratified sampling.

Figure 12. Comparison of converged EIG-based question rankings under different strategies for sampling m simulated contexts when estimating $p(a | q, I)$. The rankings are highly consistent across sampling schemes, supporting the use of uniform random sampling in the main experiments.

- **Naturalness and fluency:** Is the wording natural, fluent, conversational, and easy to understand,

Table 6. Qwen3-8B Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)						p_{ohuman} (QWK)					
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara
LLM Direct	0.7275	0.9392	0.8512	0.9167	0.8585	0.9283	0.3172	0.2606	0.4451	0.3742	0.3158	0.4350
LLM + CoT	0.6709	0.6501	0.7048	0.8685	0.7449	0.8127	0.2934	0.2667	0.4126	0.3600	0.3507	0.4734
CheckEval	0.7245	0.9261	0.8783	0.8439	0.8923	0.8945	0.1840	0.2302	0.3836	0.2767	0.2841	0.4097
Ours (EIG)	0.7545	0.9622	0.8941	0.9831	0.9203	0.9694	0.3808	0.2758	0.5124	0.4033	0.3916	0.4847

Table 7. Qwen3-30B-A3B-Instruct-2507 Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)						p_{ohuman} (QWK)					
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara
LLM Direct	0.7327	0.9237	0.7734	0.9028	0.8325	0.9024	0.3124	0.2421	0.3824	0.3545	0.3347	0.4034
LLM + CoT	0.6342	0.7241	0.6823	0.8482	0.7461	0.8427	0.3471	0.2628	0.4323	0.3943	0.3746	0.4456
CheckEval	0.7417	0.9382	0.7981	0.9313	0.8512	0.9142	0.2934	0.2273	0.3593	0.3435	0.3245	0.3841
Ours (EIG)	0.7823	0.9672	0.8455	0.9855	0.8835	0.9575	0.3832	0.2893	0.4951	0.4248	0.4035	0.4728

Table 8. Qwen3-Next-80B-A3B-Instruct Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)						p_{ohuman} (QWK)					
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara
LLM Direct	0.5509	0.4661	0.6942	0.8437	0.6423	0.6258	0.4841	0.2892	0.4507	0.4024	0.4248	0.5267
LLM + CoT	0.4638	0.3016	0.6137	0.8024	0.6126	0.5742	0.5213	0.3031	0.4751	0.4303	0.4651	0.5543
CheckEval	0.5713	0.4892	0.7236	0.8591	0.6637	0.6484	0.4529	0.2819	0.4044	0.3592	0.4034	0.4845
Ours (EIG)	0.6092	0.5426	0.7949	0.8963	0.7024	0.6852	0.5587	0.3344	0.4902	0.4894	0.4892	0.5884

Table 9. Llama-3.3-70B-Instruct Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)						p_{ohuman} (QWK)					
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara
LLM Direct	0.8127	0.9212	0.8542	0.9633	0.8272	0.9242	0.2671	0.2273	0.4118	0.3667	0.3237	0.3875
LLM + CoT	0.7904	0.8230	0.7951	0.9230	0.7731	0.8729	0.3026	0.2455	0.4572	0.3800	0.3648	0.4383
CheckEval	0.8433	0.9588	0.8626	0.9845	0.8453	0.9360	0.2248	0.2061	0.3642	0.3667	0.3039	0.3240
Ours (EIG)	0.8702	0.9755	0.8873	1.0000	0.8946	0.9607	0.3392	0.2818	0.4837	0.4176	0.3902	0.4639

Table 10. DeepSeek-R1-0528 Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)						p_{ohuman} (QWK)					
	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara	CharEval	rJokes	Love-EQ	MT_bench	TSA-MD	Capybara
LLM Direct	0.5724	0.3706	0.7451	0.9038	0.6829	0.6347	0.4719	0.2532	0.4951	0.4467	0.4127	0.5346
LLM + CoT	0.5247	0.2937	0.7193	0.8450	0.6351	0.6039	0.5135	0.2609	0.4845	0.4533	0.4412	0.5639
CheckEval	0.5912	0.3934	0.7517	0.9126	0.6933	0.6732	0.4028	0.2242	0.3927	0.4128	0.3892	0.5014
Ours (EIG)	0.6138	0.4288	0.7958	0.9364	0.7453	0.7192	0.5462	0.2939	0.5383	0.4769	0.4553	0.5933

without obvious “machine-like” stiffness?

- **Emotional attunement:** Does the reply recognize and respond to the interlocutor’s emotions and implicit signals, showing care, interest, or emotional alignment?
- **Creativity and humor:** Does the reply show creativity, wit, or humor (when appropriate), rather than being mechanical, dull, or completely off-topic?
- **Politeness and respect:** Even in teasing/flirting contexts, does the reply maintain basic respect and boundaries, avoiding clear offense or insult?

- **Context understanding:** Is the reply coherent and consistent with the prior turns, avoiding obvious non sequiturs or misunderstandings of the situation?

• **Holistic rating rubric (1–5):**

- **5 – Excellent:** Strong overall performance; highly aligned with romantic chat contexts and outstanding on most reference dimensions.
- **4 – Good:** Generally good with minor flaws; overall natural, appropriate, and reasonably engaging.
- **3 – Neutral/Average:** Mixed strengths and weaknesses; acceptable but unremarkable, or slightly

Table 11. Absolute construction cost summarized via break-even analysis (equivalent evaluation samples) and alignment performance (Avg QWK).

Method	Construction Complexity	Equivalent Eval Samples (Break-Even)	Performance (Avg QWK)
CheckEval	Rubric Only	~ 13	0.3118
Ours (EIG)	Rubric + Simulations	~ 80	0.4719

Table 12. Absolute construction cost breakdown (formatted as “#calls $\times$ tokens/call”). Our construction includes both rubric construction and simulated answers for EIG estimation.

Dataset	CheckEval (rubric construction)	Ours (rubric construction + simulated answers)
CharEval	$10 \times 1.4k$	$10 \times 1.4k + 63 \times 1.1k$
rJokes	$14 \times 1.1k$	$14 \times 1.1k + 79 \times 1.0k$
Love-EQ	$12 \times 1.3k$	$12 \times 1.3k + 62 \times 1.1k$
MT_bench	$10 \times 1.5k$	$10 \times 1.5k + 59 \times 1.1k$
TSA-MD	$12 \times 1.3k$	$12 \times 1.3k + 63 \times 1.0k$
Capybara	$16 \times 1.5k$	$16 \times 1.5k + 51 \times 1.1k$
AVG	$12.3 \times 1.35k$	$16 \times 1.35k + 62.8 \times 1.06k$

problematic on some dimensions.

- **2 – Poor:** Multiple noticeable issues (e.g., unnatural phrasing, weak empathy, lack of humor/creativity, or mildly disrespectful behavior).
- **1 – Very poor:** Largely unacceptable (e.g., severely unnatural, strongly offensive, severely misunderstanding the context, or offering almost no positive interaction value).
- **Reference materials and ambiguity handling:** Annotators are provided with (i) the full multi-turn dialogue, (ii) a brief scenario description, and (iii) standardized definitions and examples for each reference dimension. For ambiguous cases (e.g., seemingly rude wording used as friendly banter), annotators should judge based on the **overall conversational tone and the interlocutor’s reaction**, reflecting realistic flirting/teasing interactions.

Before formal annotation, all annotators completed a **calibration phase**: they rated a shared set of examples and compared outcomes against reference ratings and group discussion. Only after achieving sufficient alignment did annotators proceed to the main annotation stage.

A.8.2. ANNOTATION PROCEDURE

Following the guideline above, we conducted systematic annotation for Love-EQ. We first curated multi-turn dialogue instances from cleaned and de-identified romantic chat data, retaining samples that contain a male “**Golden**” utterance, and then applied unified formatting and noise filtering. **The “Golden” markers were initially assigned by GPT-4.1 and subsequently reviewed and filtered by human annotators**

to ensure correctness and relevance. We recruited annotators with Chinese reading proficiency and familiarity with online chat conventions, and provided standardized training and trial annotation to ensure a consistent understanding of the task and reference dimensions.

In the main annotation stage, each instance (i.e., one male Golden utterance with its full context) was independently rated by **three** annotators, each assigning a **single holistic score (1–5)** while freely weighing the reference dimensions. The final dataset contains **1,039** annotated instances. To ensure quality, we inserted a small set of validation items with relatively unambiguous expected ratings, and manually reviewed conspicuous disagreements or abnormal annotations for re-checking and potential re-annotation.

After annotation, we quantified inter-annotator agreement on the 1–5 holistic ratings using **Fleiss’ κ** , computed over the three annotators’ ratings, obtaining $\kappa = 0.7326$.

A.9. Sampling Strategies for Simulated Contexts in EIG Estimation

To estimate $p(a | q, I)$ with m simulated contexts, our main pipeline constructs each context I by **uniformly sampling** instances from the dataset D at random. This simple strategy serves as a practical approximation to the overall data distribution and is used throughout the main experiments.

To examine whether the EIG-based question ranking is sensitive to how simulated contexts are sampled, we additionally compare the above random sampling against a **cluster-then-stratified sampling** strategy. Concretely, we first cluster all instances in D (e.g., using text representations/embeddings) into multiple clusters, and then construct each simulated

Table 13. Comparison of self-consistency (κ_{self} , measured by Cohen’s κ) and human alignment (p_{ohuman} , measured by QWK). Best results are in bold, second-best are shaded.

Method	κ_{self} (Cohen’s κ)							p_{ohuman} (QWK)						
	CharEval	rJokes	Love_EQ	MT_bench	TSA_MD	Capybara	Avg	CharEval	rJokes	Love_EQ	MT_bench	TSA_MD	Capybara	Avg
LLM Direct	0.6218	0.4420	0.7924	0.9314	0.7768	0.8519	0.7361	0.4534	0.2526	0.4736	0.3864	0.3949	0.5125	0.4122
PoLL (different)	0.6317	0.4479	0.7834	0.9407	0.7714	0.8497	0.7375	0.4627	0.2417	0.4818	0.3957	0.4045	0.5025	0.4148
PoLL (same)	0.6264	0.4457	0.7947	0.9346	0.7790	0.8528	0.7389	0.4541	0.2532	0.4783	0.3869	0.3977	0.5142	0.4141
Ours (EIG)	0.6728	0.4832	0.8513	0.9523	0.8145	0.8974	0.7786	0.5186	0.3394	0.5455	0.4533	0.4367	0.5925	0.4810

Table 14. Paired bootstrap significance analysis on human alignment. Δ denotes the improvement in observed human agreement: $\Delta = p_{\text{ohuman}}^{\text{ours}} - p_{\text{ohuman}}^{\text{baseline}}$. We report 95% bootstrap confidence intervals (CI) over evaluation instances. All CIs exclude zero.

Dataset	$\Delta(\text{Ours} - \text{LLM Direct})$ (95% CI)	$\Delta(\text{Ours} - \text{LLM+CoT})$ (95% CI)	$\Delta(\text{Ours} - \text{PoLL})$ (95% CI)	$\Delta(\text{Ours} - \text{CheckEval})$ (95% CI)
CharEval	$\Delta=0.0652$, [0.0260, 0.1138]	$\Delta=0.0212$, [0.0107, 0.0303]	$\Delta=0.0559$, [0.0390, 0.0754]	$\Delta=0.1413$, [0.1220, 0.1710]
rJokes	$\Delta=0.0868$, [0.0424, 0.1317]	$\Delta=0.0333$, [0.0213, 0.0520]	$\Delta=0.0977$, [0.0670, 0.1375]	$\Delta=0.1333$, [0.1090, 0.1628]
Love-EQ	$\Delta=0.0719$, [0.0350, 0.0914]	$\Delta=0.0329$, [0.0240, 0.0441]	$\Delta=0.0637$, [0.0425, 0.0936]	$\Delta=0.1621$, [0.1325, 0.1920]
MT_Bench	$\Delta=0.0669$, [0.0560, 0.0750]	$\Delta=0.0400$, [0.0325, 0.0498]	$\Delta=0.0576$, [0.0341, 0.0778]	$\Delta=0.1406$, [0.1272, 0.1711]
TSA-MD	$\Delta=0.0418$, [0.0150, 0.0707]	$\Delta=0.0129$, [0.0030, 0.0221]	$\Delta=0.0322$, [0.0125, 0.0523]	$\Delta=0.0970$, [0.0622, 0.1268]
Capybara	$\Delta=0.0800$, [0.0640, 0.1189]	$\Delta=0.0625$, [0.0520, 0.0830]	$\Delta=0.0900$, [0.0762, 0.1105]	$\Delta=0.1000$, [0.0740, 0.1285]

context by sampling instances in a **stratified** manner across clusters (e.g., proportional allocation or fixed quota per cluster). Intuitively, this variant encourages broader coverage of diverse conversational patterns within each simulated context.

Figure 12 shows the **converged** EIG-induced question ranking under the two sampling schemes. As can be seen, the resulting rankings are **highly consistent** after convergence, with only minor local swaps among adjacent questions. Therefore, we adopt the simpler **uniform random sampling** method in the main pipeline.

A.10. Additional Baseline: Majority Voting with the Same Judge Model (PoLL same)

In addition to **PoLL (different)** reported in our main experiments (majority voting across diverse model families), we further evaluate **PoLL (same)**, where we sample multiple independent judging runs from the *same* model and aggregate the final decision via majority voting. As shown in Table 13, the effect on κ_{self} is *dataset-dependent*: PoLL (different) achieves higher self-consistency on CHAREVAL, RJOKES, and MT_BENCH, while PoLL (same) is slightly higher on LOVE_EQ, TSA_MD, and CAPYBARA. Despite these modest gains in self-consistency, both voting schemes yield only limited improvements in human alignment (p_{ohuman}) compared to **LLM Direct**. In contrast, our **EIG** method consistently delivers substantially better human alignment across all benchmarks.

A.11. Paired Significance Analysis for Human Alignment

To go beyond reporting raw human agreement (p_{ohuman}), we conduct a **paired significance analysis** at the instance level. For each dataset and each baseline method, we compute the per-item agreement indicator with human judgments and

report the paired difference $\Delta = p_{\text{ohuman}}^{\text{ours}} - p_{\text{ohuman}}^{\text{baseline}}$. We then estimate **95% confidence intervals** for Δ via **bootstrap resampling** over evaluation instances (with paired resampling to preserve the same items across methods). As shown in Table 14, the **lower bound** of every 95% CI is **strictly positive**, indicating that our improvements in human alignment are statistically significant and consistent across all datasets and baselines.

A.12. Absolute Construction Cost and Comparison with CheckEval

We report the **absolute one-time construction cost** of each method in terms of **API calls and token usage** (Table 12), where each entry is formatted as “#calls $\times$ tokens/call”. We further summarize the **break-even point** (Table 11), defined as the number of evaluation samples whose cumulative evaluation cost is equivalent to the one-time construction cost. Compared to **CheckEval**, our method incurs a higher upfront cost because it requires **rubric construction + simulated answers** for EIG estimation (about ~ 80 vs. ~ 13 equivalent samples). This additional cost is a one-time investment and yields substantially better alignment: our method improves **Avg QWK** from 0.3118 to 0.4719 (+51% relative). After construction, the **per-sample evaluation cost** of our method is comparable to CheckEval/LLM Direct, while baselines such as PoLL (multiple calls per sample) and LLM+CoT (longer generations) incur consistently higher marginal costs for every evaluated item.